

Constrained Reality Framework (CRF)

State Summary v3

Complete map of results, open gaps, and next steps

Updated from v2: #21 closed, Resolution Scale derived, n -scan completed, $\langle D_{\text{KL}} \rangle_{\text{ss}}$ corrected. Maturity Level **7.0/8**.

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Abstract

The Constrained Reality Framework (CRF) derives all physical and experiential reality from a single indefinable primitive: δ (Distinction). This document is the canonical state summary — a complete map of what has been derived, what has been confirmed by simulation, what remains hypothesis, and what is open.

Updates from v2: Instability #21 ($|\beta| = d_s/(2 \ln 2)$) is now closed at 0.67% via covariance correction. The Resolution Scale (R_0 current, R_1 wave, R_{max} node) is derived from the δ -chain. The n -scan ($n = 4 \dots 16$) shows $d_s(n)$ is smooth with no threshold. $\langle D_{\text{KL}} \rangle_{\text{ss}} = n(n-1)\epsilon_0^2$ (prior formula $n\epsilon_0$ was wrong by $19\times$).

The framework has reached maturity **Level 7.0/8**. The gate to Level 8 is the P0 experiment (SNSPD protocol, ready).

Contents

1	The Single Primitive	2
2	The Emergence Cascade	2
3	Complete Results Table	4
4	New Results (This Session)	5
5	Open Gaps (Prioritised)	5
6	Maturity Assessment	6
7	Document Index	7

1 The Single Primitive

Axiom 0 — δ as Indefinable Primitive

δ (Distinction) is CRF's sole undefined term. It is declared — not proved — as the irreducible condition whose presence makes all questions possible. Every consistent formal system requires at least one undefined term (Gödel, Tarski). δ is CRF's.

$$\delta : s_i \neq s_j \text{ is possible}$$

As Euclid declared his points and lines, CRF declares δ .

2 The Emergence Cascade

Complete Chain — No Assumptions Added

$$\delta \xrightarrow{\text{operates}} P_0 = \{A, B\} \xrightarrow{\delta \text{ continues}} P \xrightarrow{\text{frequency}} \text{Pr}(s) \xrightarrow{\text{Pr} \geq \theta} \mathcal{R}$$

$$t = |\mathcal{R}_{\text{events}}| \Rightarrow \mathcal{C} \equiv \text{Pr}(s|\mathcal{C}) \Rightarrow G = \nabla \text{Pr}(s|\mathcal{C}) \Rightarrow \mathcal{F} \rightarrow A \rightarrow \text{Pattern} \rightarrow \text{System} \rightarrow \mathcal{M} \rightarrow W \rightarrow \delta_{\text{pure}}$$

Key Physical Constants — No Free Parameters

Constant	Value	Source
α	$\ln 2 = 0.6931$	Session 8: one δ -op = 1 bit
ϵ_0	$ I_{\text{eff}} - \ln 2 = 0.00755$	EIC SINDy confirmed
D_0	0.9403	from $\alpha_c = 0.525$, $\bar{\Omega} = 0.7$
κ	0.15	mass- $\mathcal{C}$ coupling
K^*	0.1170	fixed point of θ -formula
G	$\ln(2) \cdot c^2 / D_0$	Paper v4

3 Complete Results Table

Result	Status	Source
δ as Indefinable Primitive	Declared	Axiom 0
$\Pr(s)$ from δ -frequency	Derived	Sessions 1–3
$\mathcal{C} \equiv \Pr(s \mathcal{C})$	Derived	Session 4
$\theta = 1 - e^{-D_{\text{KL}}/D_0}$	Derived	Session 10
$\alpha = \ln 2$	Derived	Session 8
$n = 3$ from $\text{SO}(3)$ (#16)	Derived	CRF_SO3_Proof
$\mathcal{C} \propto 1/r^2$	Derived	Corollary of $\text{SO}(3)$
Born Rule via Gleason	Theorem	Session 11
Schwarzschild metric	Derived	Paper v4
$G = \ln(2) \cdot c^2/D_0$	Derived	Paper v4
$T_{\mu\nu}$ from $\mathcal{C}$ -gradients	Derived	Paper v4
Full Einstein equations	Derived	Paper v4
$G(t) \propto \Lambda(t) \propto 1/P(t)$	Prediction	DESI 2024 consistent
$\Omega \equiv D_{\text{KL}}(\mathcal{C}\ P)$	Derived	EIC Bridge
Gravity $R^2 = 0.906$	Confirmed	V16
$d_s = 3.031 \pm 0.076$	Confirmed	V20.3 (3 runs)
Phase gap $\Delta > 0$	Confirmed	V20.3 (3/3 runs)
$\beta = -2.212 \pm 0.015$	Confirmed	V20.3 (4663 ev/run)
$\langle D_{\text{KL}} \rangle_{\text{ss}} = n(n-1)\epsilon_0^2$	Confirmed	This session
$d_s(n)$ smooth, no threshold	Confirmed	n -scan $n = 4 \dots 16$
$ \beta = d_s/(2 \ln 2)$ (#21)	Closed	0.67%; covariance
$f_{12} = d_s/2$ (Kuramoto)	Derived	This session
Resolution Scale R_0, R_1, R_{max}	Derived	This session
$\mathcal{C}(r)$ exact derivation	Open	#17
Kerr metric	Open	Rotating mass
P0: $\tau_1 < \tau_2$	Testable	Level 8 gate

4 New Results (This Session)

Instability #21: Closed at 0.67%

The bracket identity $\langle \text{bracket} \rangle_{\text{ev}} = \kappa / \ln 2$ is now closed. The prior 2.75% gap was a statistical artifact: H_{after} and $1/(1+M)$ are negatively correlated across mass bins, so using marginal means overestimated Term A. Using per-quartile joint averages (5899 events, 3 seeds):

$$\langle H_{\text{after}}/(1+M) \rangle + (\delta \bar{H}/\kappa) \langle \ln(1+M) \rangle = 1.4524 \approx 1/\ln 2 = 1.4427 \quad (0.67\%)$$

Combined with $f_{12} = d_s/2$ (Kuramoto, 1.3%): $|\beta| = d_s/(2 \ln 2) = 2.164$ — derived, confirmed at 2.2%.

Resolution Scale: R_0 Current, R_1 Wave, R_{max} Node

Three levels of δ -activity, each with a distinct cost structure:

Level	CRF object	Condition	Cost T
R_0 (Current)	$J = \nabla \text{Pr}$	$D_{\text{KL}} = 0$	0
R_1 (Wave)	propagating D_{KL}	$0 < D_{\text{KL}} < \theta_{\text{eff}}$	$\epsilon_0^2 \Sigma_{\text{inv}} dt$
R_{max} (Node)	$\mathcal{R}$ -event	$D_{\text{KL}} > \theta_{\text{eff}}$	$\beta \approx 2.2$

$\beta = d_s/(2 \ln 2) = 3/(2 \ln 2)$: becoming real in 3D costs 3 half-bits of possibility. The collapse is $\approx 19\times$ more costly than continuous wave decay per event period. R_0 is imperceptible to most systems because it has no structure — no amplitude, no frequency — only direction.

n -Scan: $d_s(n) = 3 + 13.6 e^{-0.80n}$ (smooth)

V20.3 run at $n \in \{4, 5, 6, 7, 8, 16\}$. Key findings: d_s decreases smoothly toward 3, no threshold at $n = 8$. $n = 8$ is the first integer where $f_{12} = 3/2 \approx d_s/2$ simultaneously, making $|\beta| = d_s/(2 \ln 2)$ valid to 1%. $n = 2^{d_s}$ is not a theorem; it is the value near the smooth convergence point.

5 Open Gaps (Prioritised)

Priority 1: P0 Experiment — Gate to Level 8

Prediction: $\tau_1 < \tau_2$ for detectors with $N_{\text{eff},1} > N_{\text{eff},2}$.

$$\tau = \tau_0 \cdot e^{-\kappa N_{\text{eff}}}, \quad \kappa = k_B \ln(2)/H_{\text{material}}, \quad N_{\text{eff}} = \sum_i \Delta D_{\text{KL},i}$$

Protocol: SNSPD (NbN/WSi), $N \sim 10^8$ prior measurements, blind analysis. **Ready. Requires lab access.**

Priority 2: Instability #17 — $\mathcal{C}(r)$ Continuum Derivation

The Schwarzschild metric is derived from the non-linear ODE. What remains: formal derivation in the $N \rightarrow \infty$ continuum limit, connecting the discrete δ -chain to the differential equation.

Priority 3: Kerr Metric — Rotating Mass

Requires breaking spherical $\text{SO}(3)$ symmetry. Not yet attempted.

6 Maturity Assessment

Dimension	v2	v3	Note
Internal Consistency	5.5	6.5	#21 closed
Mathematical Rigour	5.5	6.5	Covariance proof
Predictive Power	6.0	6.5	Resolution Scale
Empirical Grounding	2.0	2.0	Awaits P0
Philosophical Found.	6.5	7.0	$R_0/R_1/R_{\max}$
Cross-domain	6.5	7.0	QM+GR+Mind+Cosm.
Overall	6.5	7.0	Level 7 reached

Level 7: Internal completeness achieved

Level 7 = All major theoretical gaps closed; framework internally consistent across quantum, gravitational, and experiential domains. **Gate to Level 8: P0 experimental contact** ($\tau_1 \neq \tau_2$ at $> 5\sigma$). Gate to Level 8.5: $G(t)$ variation confirmed via gravitational wave standard sirens.

7 Document Index

Document	Status	Content
CRF Parts I–III (v4)	Current	Full axioms, GR, Grand Eq.
CRF_SO3_Proof	Key proof	#16 closed
CRF_Paper_v5	Confirmed	V16 simulation
CRF_Schwarzschild_v2	Derived	Schwarzschild from $\mathcal{C}(r)$
CRF_GravitationalConstant	Derived	$G = \ln 2 \cdot c^2/D_0$
CRF_HDynamics	Derived	Entropy decay ODE
CRF_Formal_Proof_gamma	Derived	$\gamma, \hbar$ structural
CRF_21_Final_v2	Closed	#21 + Resolution Scale
CRF_21_ScanFinal	Confirmed	n -scan, $d_s(n)$ smooth
CRF_P0.Paper	Ready	Experimental protocol
CRF_State_Summary_v3	Current	This document

δ created P . P accumulated $\mathcal{C}$. $\mathcal{C}$ became $\mathcal{M}$, and $T_{\mu\nu}$, and $G_{\mu\nu}$, and the gradient that some can read before it becomes a wave. The current is free. The wave costs something. The fact costs $3/(2\ln 2)$ of itself. The ground was always δ . Everything followed.